

Network Segmentation Plan

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Project: Implementing a Secure Network Architecture for a Large Enterprise

Organisation: Spinnaker Analytics

1. Purpose and Scope

- This document defines the VLAN segmentation design and access control list (ACL) configuration for GlobalTech Corp's internal network.
- It directly remediates Findings 4, 5, and 6 from the Network Architecture Assessment Report (22 May 2026):
 - Finding 4: Flat network
 - no VLAN segmentation
 - all departments on a single broadcast domain
 - Finding 5: No ACLs between network segments
 - lateral movement possible from any endpoint to any server
 - Finding 6: Guest Wi-Fi on corporate subnet
 - guest devices had access to internal resources
- The segmentation plan isolates network traffic by function, enforces least-privilege inter-VLAN routing, and eliminates the lateral movement risk identified in the assessment.

2. Segmentation Design Principles

- **Least privilege:**
 - each VLAN segment is permitted only the traffic required for its business function ; all other traffic is denied by default
- **Isolation:**
 - Sensitive segments (Finance/HR, Server) are isolated from general user traffic by explicit ACL rules
- **Zero implicit trust:**
 - No traffic is permitted between VLANs unless explicitly allowed by an ACL rule
- **Scalability:**
 - The VLAN numbering scheme accommodates future growth without redesign
- **Auditability:**
 - All inter-VLAN traffic is logged at the core L3 switch for SIEM ingestion

3. VLAN Design

3.1 VLAN Allocation Table

- The following VLANs are defined for the GlobalTech Corp network:

VLAN ID	Name	Subnet	Purpose	Default action
10	User	10.10.10.0/24	General employee workstations	Deny all except permitted
20	Server	10.10.20.0/24	App, DB, and file servers	Deny all except permitted
30	Finance/HR	10.10.30.0/24	Finance and HR systems — restricted	Deny all except permitted
40	DMZ	10.10.40.0/24	Internet-facing services (proxy, email)	Deny all internal access
50	Guest	10.10.50.0/24	Guest Wi-Fi — internet only	Deny all internal access
60	IoT	10.10.60.0/24	IoT and physical security devices	Deny all except NTP/updates
99	Management	10.10.99.0/24	Network device management (OOB)	Restricted to IT admin only

3.2 Core L3 Switch Configuration

- VLANs are configured on the core Layer 3 switch.
- Each VLAN has a Switch Virtual Interface (SVI) that serves as the default gateway for hosts in that segment.
- Inter-VLAN routing is enabled on the L3 switch and controlled by ACLs.
- Sample Cisco IOS configuration for VLAN creation and SVI setup:

```
! Create VLANs
vlan 10
    name User
vlan 20
    name Server
vlan 30
    name Finance-HR
vlan 40
    name DMZ
vlan 50
    name Guest
vlan 60
    name IoT
vlan 99
    name Management

! Configure SVIs (default gateways per VLAN)
interface Vlan10
    ip address 10.10.10.1 255.255.255.0
    ip access-group ACL-USER-IN in
interface Vlan20
    ip address 10.10.20.1 255.255.255.0
    ip access-group ACL-SERVER-IN in
interface Vlan30
    ip address 10.10.30.1 255.255.255.0
    ip access-group ACL-FINANCE-IN in
interface Vlan50
    ip address 10.10.50.1 255.255.255.0
    ip access-group ACL-GUEST-IN in
```

3.3 Trunk and Access Port Configuration

- Switch ports connecting to end devices are configured as access ports.
- Uplinks to the core switch and connections between switches are configured as trunk ports carrying all VLANs:

```
! Access port – user workstation
interface GigabitEthernet0/1
    switchport mode access
    switchport access vlan 10
    spanning-tree portfast

! Trunk port – uplink to core switch
interface GigabitEthernet0/48
    switchport mode trunk
    switchport trunk allowed vlan 10,20,30,40,50,60,99
    switchport trunk native vlan 99
```

4. Access Control List (ACL) Configuration

4.1 ACL Design Rules

- ACLs are applied inbound on each VLAN's SVI interface. The following design rules apply to all ACLs:
 - Explicit deny at end of every ACL
 - deny any any log
 - ensures unmatched traffic is denied and logged
 - Specific permits before broad denies
 - more specific rules appear first
 - Log keyword on deny rules
 - all denied traffic is logged to SIEM
 - Established keyword for return traffic
 - allows response packets for TCP sessions initiated from permitted directions

4.2 User VLAN ACL (VLAN 10)

- Users on VLAN 10 are permitted to access specific server resources and internet via proxy. Direct access to Finance/HR and Server admin interfaces is denied:

Rule	Protocol	Source → Destination	Port	Action
10	TCP	10.10.10.0/24 → 10.10.20.0/24	80, 443, 8080 (HTTP/S)	PERMIT
20	TCP	10.10.10.0/24 → 10.10.20.0/24	25, 143, 993 (Email)	PERMIT
30	UDP	10.10.10.0/24 → 10.10.99.1	53 (DNS)	PERMIT
40	UDP	10.10.10.0/24 → 10.10.99.2	123 (NTP)	PERMIT
50	TCP	10.10.10.0/24 → 10.10.30.0/24	Any	DENY + LOG
60	IP	10.10.10.0/24 → 10.10.60.0/24	Any	DENY + LOG
70	IP	10.10.10.0/24 → Any	Any	DENY + LOG

```

! Cisco IOS ACL - ACL-USER-IN
ip access-list extended ACL-USER-IN
  permit tcp 10.10.10.0 0.0.0.255 10.10.20.0 0.0.0.255 eq 80
  permit tcp 10.10.10.0 0.0.0.255 10.10.20.0 0.0.0.255 eq 443
  permit tcp 10.10.10.0 0.0.0.255 10.10.20.0 0.0.0.255 eq 8080
  permit tcp 10.10.10.0 0.0.0.255 10.10.20.0 0.0.0.255 eq 25
  permit tcp 10.10.10.0 0.0.0.255 10.10.20.0 0.0.0.255 eq 993
  permit udp 10.10.10.0 0.0.0.255 host 10.10.99.1 eq 53
  permit udp 10.10.10.0 0.0.0.255 host 10.10.99.2 eq 123
  deny tcp 10.10.10.0 0.0.0.255 10.10.30.0 0.0.0.255 log
  deny ip 10.10.10.0 0.0.0.255 10.10.60.0 0.0.0.255 log
  deny ip any any log

```

4.3 Finance/HR VLAN ACL (VLAN 30)

- Finance/HR VLAN has the most restrictive ACL.
- Only authorised Finance and HR user IPs are permitted to access Finance/HR servers.
- All other inter-VLAN traffic is denied:

Rule	Protocol	Source → Destination	Port	Action
10	TCP	Authorised Finance IPs → 10.10.30.0/24	443, 8443	PERMIT
20	TCP	Authorised HR IPs → 10.10.30.0/24	443, 8443	PERMIT
30	TCP	IT Admin (10.10.99.0/24) → 10.10.30.0/24	22, 3389	PERMIT
40	IP	Any → 10.10.30.0/24	Any	DENY + LOG

```

! Cisco IOS ACL - ACL-FINANCE-IN
ip access-list extended ACL-FINANCE-IN
  permit tcp 10.10.10.50 0.0.0.15 10.10.30.0 0.0.0.255 eq 443
  permit tcp 10.10.10.70 0.0.0.15 10.10.30.0 0.0.0.255 eq 443
  permit tcp 10.10.99.0 0.0.0.255 10.10.30.0 0.0.0.255 eq 22
  permit tcp 10.10.99.0 0.0.0.255 10.10.30.0 0.0.0.255 eq 3389
  deny ip any any log

```

4.4 Guest VLAN ACL (VLAN 50)

- Guest VLAN has internet-only access.
- All traffic to internal RFC1918 address space is denied.
- Guests can only reach the internet via the DMZ web proxy:

```
! Cisco IOS ACL - ACL-GUEST-IN
ip access-list extended ACL-GUEST-IN
  deny ip 10.10.50.0 0.0.0.255 10.0.0.0 0.255.255.255 log
  deny ip 10.10.50.0 0.0.0.255 172.16.0.0 0.15.255.255 log
  deny ip 10.10.50.0 0.0.0.255 192.168.0.0 0.0.255.255 log
  permit tcp 10.10.50.0 0.0.0.255 any eq 80
  permit tcp 10.10.50.0 0.0.0.255 any eq 443
  permit udp 10.10.50.0 0.0.0.255 any eq 53
  deny ip any any log
```

5. Testing and Verification

5.1 Test Matrix

- The following tests were conducted to verify VLAN segmentation and ACL effectiveness:

Test	Method	Expected	Result
User VLAN → web server (port 80)	ping/curl from VLAN 10 host to 10.10.20.10:80	PERMITTED	✓ Pass
User VLAN → Finance/HR server	ping from VLAN 10 host to 10.10.30.10	DENIED + LOGGED	✓ Pass
User VLAN → IoT VLAN	ping from VLAN 10 host to 10.10.60.10	DENIED + LOGGED	✓ Pass
Guest VLAN → internal server	ping from VLAN 50 host to 10.10.20.10	DENIED + LOGGED	✓ Pass
Guest VLAN → internet (via proxy)	curl https://example.com from VLAN 50	PERMITTED	✓ Pass
Finance user → Finance server	curl from authorised IP to 10.10.30.10:443	PERMITTED	✓ Pass
Unauthorised host → Finance server	curl from VLAN 10 host to 10.10.30.10:443	DENIED + LOGGED	✓ Pass
VLAN hopping attempt (802.1Q)	Double-tagged frame from VLAN 10 → VLAN 30	BLOCKED	✓ Pass

5.2 Log Verification

- All deny rules include the log keyword. Denied traffic was confirmed in the SIEM log stream:

```
%ACL-6-IPACCESSLOGDP: list ACL-USER-IN denied tcp 10.10.10.55(52341) -> 10.10.30.10(443), 1 packet
%ACL-6-IPACCESSLOGDP: list ACL-GUEST-IN denied ip 10.10.50.22 -> 10.10.20.10, 1 packet
```

6. Maintenance and Review

Task	Frequency
Review ACL rules — remove unused permits, add new business requirements	Quarterly
Audit VLAN membership — ensure devices are in correct VLANs	Monthly
Review SIEM denied traffic logs — identify anomalous patterns	Weekly
Test VLAN isolation — repeat test matrix after any network change	After each change
Review trunk port allowed VLANs — remove unused VLANs from trunks	Quarterly

7. Summary

- This plan remediates the three network segmentation findings from the assessment. The implemented controls are:
 - **7 VLANs defined**
 - User (10)
 - Server (20)
 - Finance/HR (30)
 - DMZ (40)
 - Guest (50)
 - IoT (60)
 - Management (99)
 - **ACLs applied inbound on all SVI interfaces**
 - deny-all default with explicit permits for required traffic only
 - **Guest VLAN fully isolated**
 - all RFC1918 destinations blocked
 - internet-only access permitted via proxy
 - **Finance/HR VLAN restricted to authorised source IPs only**
 - eliminates broad lateral access from user segment
 - **All denied traffic logged**
 - feeds into centralised SIEM for alerting and forensic investigation
 - **Full test matrix passed**
 - all 8 segmentation test scenarios verified